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Integrating Artificial Intelligence in Engineering Design: A Paradigm Shift

Satyaprasad Jadhav, Senior Specialist, ADNOC Group Projects & Engineering

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Abstract

Businesses in energy face persistent pressure to maximize return on assets while reducing project delivery cost and schedule. This paper examines how artificial intelligence (AI) with emphasis on large language models and pattern-recognition workflows—can optimize conventional engineering design processes and accelerate engineering execution across the project lifecycle. Building on qualitative interviews, case studies with consultants/contractors, and quantitative assessment of pilot deliverables, we describe an AI Engineering Design approach that: (i) extracts governed knowledge from an engineering standards library; (ii) uses pre-trained models for content and pattern recognition within typical deliverables; and (iii) operationalizes design logic through structured input forms to ensure conformance to ADNOC Group Engineering Standards. Proof-of-concept (PoC) activities indicate potential benefits on cost and schedule, with #10% optimization in engineering deliverable development and 2–4 months reduction in schedule for selected scopes, subject to data readiness and integration with existing CAD/CAE/simulation tools. We discuss key enablers—data standardization, interfaces to incumbent toolchains, and model-training resources—and provide a maturity roadmap, risk mitigations, and governance controls. We conclude that AI in engineering design is viable and timely for energy operators when deployed at procedural level on well-bounded tasks, supported by robust data stewardship and change management.

Introduction

Large, multi-disciplinary capital projects in the energy industry face a paradox: design organizations are asked to deliver more certainty (in scope, quality, cost, and schedule) with fewer resources and under tighter timelines. Design cycles are elongated by the iterative nature of engineering, the distribution of work across global centers, and the friction introduced by format and system heterogeneity—CAD drawings, equipment datasheets, line lists, specifications, calculation notes, and vendor documents all coexist, each with different fidelity and metadata. Expectations for standards compliance and auditability are rising as regulators and owners emphasize assurance and traceability. Within this environment, AI-assisted engineering promises a step-change in productivity, but practical adoption requires specificity. Rather than attempting to automate engineering end-to-end, successful implementations identify stable, repeatable, and governed procedures inside each discipline workflow where AI can add value without undermining engineering judgment. Examples include extracting structured attributes from semi-structured vendor PDFs, pre-populating equipment datasheets from standards and prior projects, flagging non-conformances against rulebooks, proposing consistent annotation templates for P&IDs, and drafting sections of the Basis of Design that are primarily templated and standards-driven. These are procedural-level insertions: bounded, reviewable, and auditable. The proposed AI Engineering Design (AI-ED) approach leverages LLMs and pattern recognition to shorten decision cycles, reduce rework, and improve standards conformance by making codified knowledge and structured inputs first-class citizens in the workflow. Engineering authority remains with subject matter experts (SMEs); AI outputs are recommendations or drafts, always routed through change-tracking and approvals.

Statement of Theory and Definitions

Definitions

AI Engineering Design (AI-ED): Application of machine learning and generative AI to produce, review, or validate engineering deliverables within a governed design process.

Procedural-level integration: Targeting specific steps within discipline workflows (e.g., P&ID assistance, datasheet checks) rather than wholesale automation.

Standards-aware generation: Constraining outputs to logic mapped to ADNOC Group Engineering Standards; users supply structured inputs to ensure traceability and compliance.

Assumptions and Constraints

Data standardization and curation are prerequisites: documents, drawings, and specifications must be normalized where practical and enriched with metadata to enable robust retrieval and rule checking. Interoperability with CAD/CAE/simulation is required to preserve a single source of truth and avoid manual re-keying; APIs or scripted connectors should bridge model and document repositories. Model training and evaluation require compute resources; control gates must ensure data classification, approval workflows, and audit trails for all AI-generated content.

Problem Framing

The central question is not whether AI can design plants, but where AI can reliably reduce waste—time lost in searching, reformatting, checking, and reconciling—without eroding engineering intent. By decomposing workflows into procedures with clear inputs, rules, and outputs, AI-ED positions AI as a force multiplier for SMEs, especially in high-volume tasks and first-pass drafting and checks.

Methods, Procedures, and Processes

Research Design

A mixed-methods approach triangulates qualitative interviews with practitioners, focused PoCs on representative deliverables, and quantitative analysis of cycle times, rework, non-conformance, and cost drivers. This design provides contextual insight and empirical evidence for decision-makers and discipline authorities.

AI-ED Workflow

The workflow comprises five phases: (1) Knowledge Extraction from a governed standards library; (2) Model Preparation using pre-trained language/vision models; (3) Standards Logic Encoding as validation rules and prompts with structured forms; (4) Generation & SME Review with change-tracking; and (5) Handover & Archive with versioning and auditable reports.

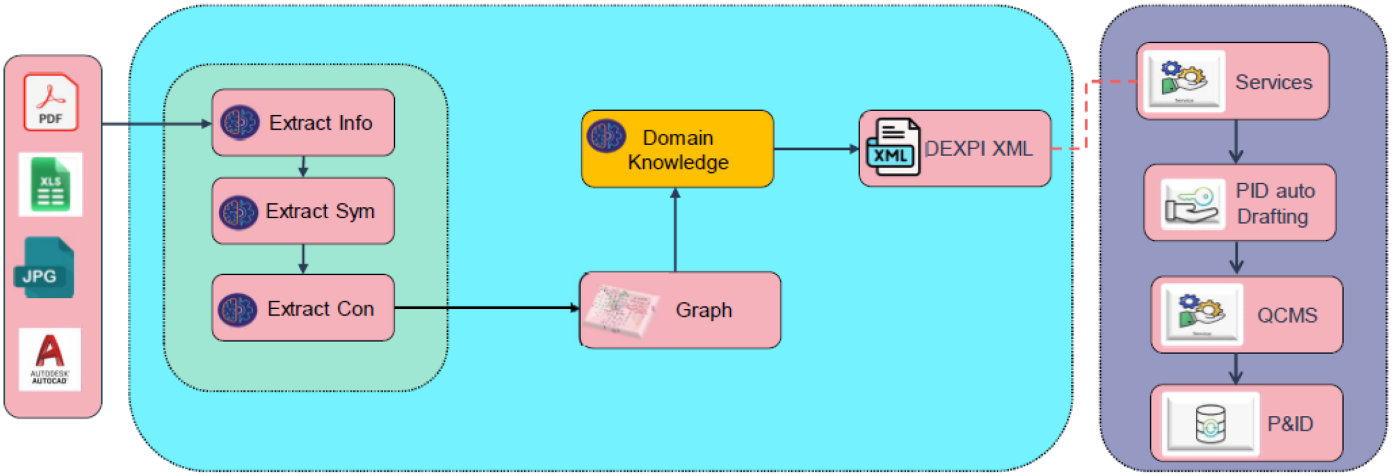


Figure 3.1—P&ID Development

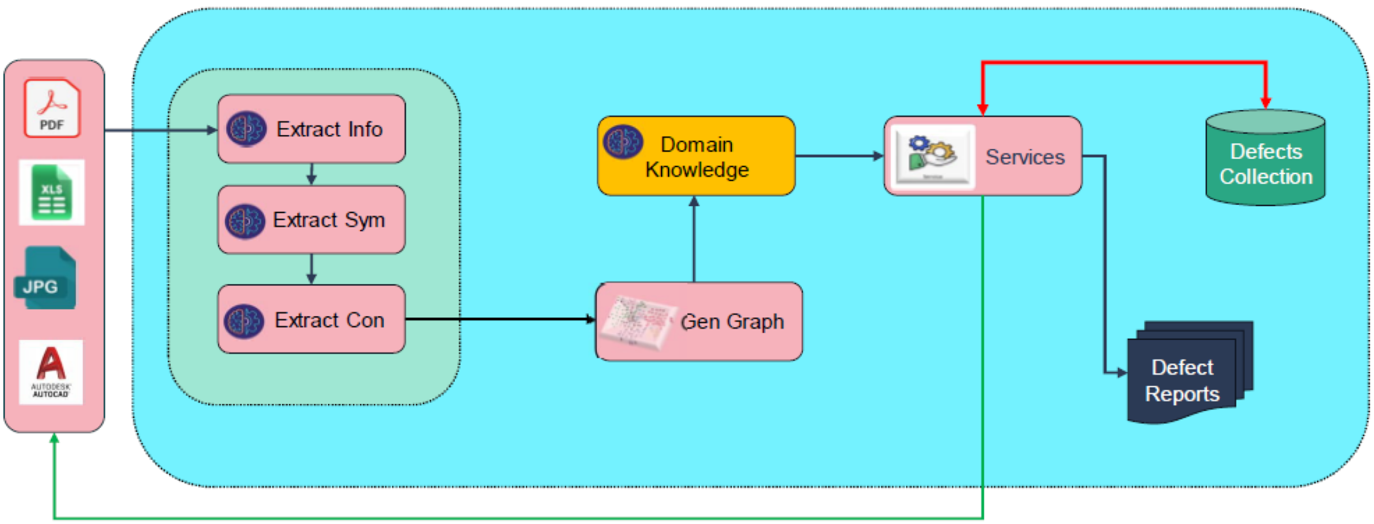


Figure 3.2—Datasheet

Knowledge and Standards Layer

A governed standards library forms the foundation—the authoritative source for design rules, datasheet templates, allowable ranges, units, and terminology. The library should be modularized, version-controlled, and metadata-rich. An extraction pipeline converts PDFs/DOCs into structured artifacts while preserving traceability back to source clauses.

Input Capture and Context Packaging

AI-ED relies on structured input forms to reduce ambiguity. Forms capture the minimal yet sufficient set of attributes for each equipment class or procedure. Inputs are validated upfront against reference tables; the resulting context bundle is fed to the model together with relevant rule excerpts, prior exemplars, and format templates.

Model Orchestration

A controller service orchestrates retrieval of standards snippets and exemplars, LLMs for text/narrative generation, vision models for pattern recognition/annotation, rule engines for deterministic checks, and post-processors that format outputs (datasheet tables, CAD layer annotations, BoD sections).

CAD/CAE/Simulation Integration

Integration should be non-intrusive and bidirectional: from CAD to AI-ED (export geometry or drawing metadata to drive checks) and from AI-ED to CAD (structured change suggestions or annotation layers for engineer acceptance). Simulation tools exchange parameter sets for sensitivity runs in higher maturity phases.

Governance, Auditability, and Controls

A governance envelope ensures safety and compliance: standards rulebooks, prompt libraries, approval matrices, data classification/retention, model accuracy/safety reviews, and document control workflows that record every AI suggestion, SME decision, and rationale. Outputs are watermarked with provenance.

MLOps and Lifecycle Management

The MLOps stack covers data versioning, model registries, automated evaluation on discipline benchmarks, gated promotion to production, and continuous monitoring. Drift alerts trigger SME review when behavior diverges from baselines, particularly after standards updates or new equipment classes.

Description and Application of Processes

The practical value of AI-assisted engineering design (AI-ED) becomes most evident when applied to **high-volume, standards-driven deliverables** that consume significant engineering effort yet follow predictable patterns. This section elaborates on five representative deliverable categories—datasheets, P&IDs, and Basis-of-Design (BoD) narratives—detailing the opportunities, workflows, and benefits of AI integration.

P&ID Development

Context and Challenges: Piping and Instrumentation Diagrams (P&IDs) are critical for defining process flow, control schemes, and safety systems. Maintaining consistency across dozens or hundreds of P&IDs is challenging, especially when multiple teams work in parallel. Common issues include inconsistent tag formats, missing safety notes, and discrepancies between P&IDs and line lists.

AI-ED Opportunity: AI-ED enhances P&ID workflows through **vision-based recognition** and **rule-driven validation**. Computer vision models detect symbols, line numbers, and instrument bubbles, while LLMs cross-reference these elements against master indices and standards. The system can also suggest annotations, such as mandatory safety notes for hazardous services, and flag missing components like isolation valves.

Workflow:

1. **Drawing Ingestion:** P&ID files are uploaded or accessed via CAD connectors.
2. **Symbol Detection:** Vision models identify equipment, lines, and instruments.
3. **Cross-Validation:** Detected elements are compared against line lists and standards.
4. **Annotation Suggestions:** AI proposes standardized notes and legends, referencing applicable rules.
5. **Discrepancy Reporting:** Missing or inconsistent elements are flagged for SME review.
6. **Integration:** Approved annotations are pushed back to the CAD environment as structured layers.

Benefits:

- **Consistency:** Uniform application of legends, notes, and tag formats across all P&IDs.
- **Error Reduction:** Early detection of discrepancies reduces downstream rework.
- **Efficiency:** SMEs focus on critical design decisions rather than repetitive checks.

Datasheets (Equipment and Instrumentation)

Context and Challenges: Datasheets are foundational documents in engineering projects, capturing critical design and operational parameters for equipment and instruments. They serve as the primary reference for procurement, fabrication, and commissioning. However, creating and maintaining datasheets is labor-intensive due to the need to reconcile multiple data sources—process design inputs, vendor documents, and corporate standards. Errors or inconsistencies in datasheets can propagate downstream, leading to costly rework or procurement delays.

AI-ED Opportunity: AI-ED addresses these challenges by automating both first-pass population and intelligent review of datasheet fields using structured inputs and standards-aware logic. The system retrieves relevant clauses from the governed standards library, applies deterministic rules for allowable ranges and units, and leverages historical project data for contextual recommendations. For example, when specifying a centrifugal pump, the AI engine can pre-fill attributes such as design pressure, temperature, and material class based on ADNOC standards and prior similar cases.

Beyond generation, AI-ED performs automated review of completed or vendor-submitted datasheets. It checks for:

- **Standards compliance:** Ensures all fields meet ADNOC Group Engineering Standards.
- **Completeness:** Flags missing mandatory fields or inconsistent units.
- **Cross-references**: Validates alignment with related documents such as P&IDs and line lists.
- **Vendor deviations**: Highlights discrepancies between vendor data and project specifications.

Workflow:

1. Structured Input Capture: Engineers provide essential parameters (e.g., service, flow rate, pressure) via a controlled form.
2. Contextual Retrieval: AI retrieves applicable standards and previous datasheet templates for similar equipment.
3. Draft Generation: The model generates a complete datasheet, highlighting any assumptions or missing inputs.
4. Automated Review: AI scans the datasheet (including vendor submissions) for compliance, completeness, and cross-document consistency.
5. Validation Report: A structured report lists all flagged issues with references to the governing standards.
6. SME Review and Approval: Engineers review AI findings in a side-by-side interface, accepting or overriding recommendations.
7. Controlled Export: Approved datasheets and review reports are archived in the document control system with full provenance metadata.

Benefits:

- Time Savings: Significant reduction in drafting and review cycles, particularly for repetitive equipment classes.
- **Quality Improvement:** Lower incidence of non-conformance due to embedded standards logic and automated checks.
- **Auditability:** Complete traceability of AI suggestions, SME decisions, and review findings for compliance audits.

Basis-of-Design (BoD) Narrative Sections

Context and Challenges: The Basis-of-Design document articulates the design philosophy, standards references, and environmental considerations for a project. While critical for governance, many sections are highly templated and repetitive, making them ideal candidates for AI assistance.

AI-ED Opportunity: AI-ED leverages LLMs to generate **first-pass drafts** of BoD sections based on project metadata and standards excerpts. For example, environmental data and design codes can be automatically inserted into relevant sections, while placeholders for project-specific details are clearly marked for SME input.

Workflow:

1. **Metadata Collection:** Project parameters (e.g., location, design codes) are captured via structured forms.
2. **Content Generation:** AI drafts narrative sections, citing applicable standards.
3. **SME Review:** Engineers refine the draft, adding project-specific details and clarifications.
4. **Version Control:** Approved sections are stored with full provenance for future reference.

Benefits:

- **Time Efficiency:** Reduces drafting time for repetitive sections.
- **Consistency:** Ensures alignment with corporate standards and terminology.
- **Audit Readiness:** Provides traceable links between BoD content and governing standards.

Summary of Impact Across Deliverables

Across these 3 categories, AI-ED consistently delivers three core benefits:

- **Efficiency:** Significant reduction in drafting and review cycles.
- **Quality:** Improved compliance with standards and reduced rework.
- **Traceability:** Comprehensive audit trails for all AI-assisted outputs.

When scaled across an entire project, these benefits translate into measurable cost savings and schedule compression, reinforcing the business case for AI adoption in engineering design.

Presentation of Data and Results

Data Readiness

The foundation of any AI-assisted engineering design (AI-ED) initiative is **data readiness**. Without structured, high-quality data, even the most advanced AI models will produce inconsistent or unreliable outputs. Success depends on implementing **content normalization** across all engineering deliverables and reference materials. This includes ensuring **consistent naming conventions**, standardized **units of measure**, and harmonized **metadata schemas** across disciplines and systems.

A critical enabler is the development of a **lightweight taxonomy** that maps engineering entities to their attributes. For example, an equipment class such as "Centrifugal Pump" should have a predefined attribute set—flow rate, head, material, design code—each with allowable ranges and units. This taxonomy becomes the backbone for **retrieval-augmented generation (RAG)** and deterministic rule checks, enabling AI models to operate within well-defined boundaries.

Equally important is the enforcement of **ID policies** for tags, documents, and data objects. Unique identifiers for lines, instruments, and equipment ensure traceability across P&IDs, datasheets, and 3D models. These IDs should be embedded in both structured registers and unstructured documents through metadata tagging, allowing AI systems to reconcile information across multiple sources without ambiguity.

Data readiness also involves **curation and cleansing** of historical project data. Legacy documents often contain inconsistencies, outdated standards references, or incomplete metadata. A structured remediation program—covering document digitization, OCR validation, and metadata enrichment—should precede AI deployment. This upfront investment reduces the risk of propagating errors and maximizes the reliability of AI-driven recommendations.

Finally, **data governance policies** must define ownership, classification, and retention rules. Given ADNOC's "Need-To-Know" classification requirements, sensitive engineering data should remain within controlled environments, with access managed through role-based permissions. These measures not only safeguard intellectual property but also ensure compliance with internal and regulatory mandates.

APIs and Connectors

Seamless integration between AI services and existing engineering tools is essential to avoid **format friction** and redundant manual steps. AI-ED should not operate as a standalone silo; instead, it must function as an **interoperable layer** within the broader engineering ecosystem.

Scripted connectors and APIs form the backbone of this interoperability. These connectors enable **bidirectional data exchange** between AI engines and core applications such as CAD, CAE, simulation tools, and document control systems. For example, when a P&ID is updated in the CAD environment, the connector should trigger an event that updates the AI-ED context, ensuring that subsequent datasheet validations or line list reconciliations reflect the latest design state.

Conversely, when AI-ED generates a recommendation—such as a revised valve specification or an annotation for a P&ID—it should be delivered back to the engineering tool in a **structured, machine-readable format** (e.g., XML or JSON), allowing engineers to accept or reject changes within their native environment. This approach preserves the **single source of truth** and eliminates the need for manual re-keying, which is a common source of errors and inefficiencies.

To maintain robustness, these APIs should be designed with **version control, error handling**, and **security protocols** such as token-based authentication and encrypted data transfer. Additionally, connectors should log all transactions for auditability, creating a transparent trail of data exchanges between systems.

Document Control and Versioning

In an AI-enabled workflow, **traceability and auditability** are non-negotiable. Every AI-assisted artifact—whether a datasheet, P&ID annotation, or Basis-of-Design narrative—must carry a **provenance record** that captures:

- **Input form hash**: A unique identifier representing the structured inputs provided by the user.
- **Standards snapshot version**: The exact version of the ADNOC Group Engineering Standards applied during generation.
- **Model version**: The identifier of the AI model used, including its training baseline and any fine-tuning iterations.
- **SME decision log**: A record of all human interventions—what was accepted, modified, or rejected—along with timestamps and user credentials.

These elements should be embedded in the document's metadata and mirrored in the document control system. Furthermore, **standards-check reports**—detailing compliance validations and flagged deviations—should be archived alongside the final deliverable. This ensures that, during audits or incident investigations, the organization can reconstruct the decision-making process and demonstrate adherence to governance protocols.

Versioning policies must also account for **iterative updates**. When standards are revised or models are retrained, the system should trigger alerts for impacted deliverables and provide a controlled mechanism for revalidation. This proactive approach prevents latent non-conformances and maintains alignment with evolving regulatory and corporate requirements.

Results

PoC Outcomes: (a) #10% reduction in engineering deliverable development effort for tested scopes; (b) potential 3–6 months project schedule reduction from accelerated engineering cycles; (c) key enablers include data structuring/standardization, robust interfaces with CAD/simulation, and adequate resources for training and evaluation.

PoC Results Table

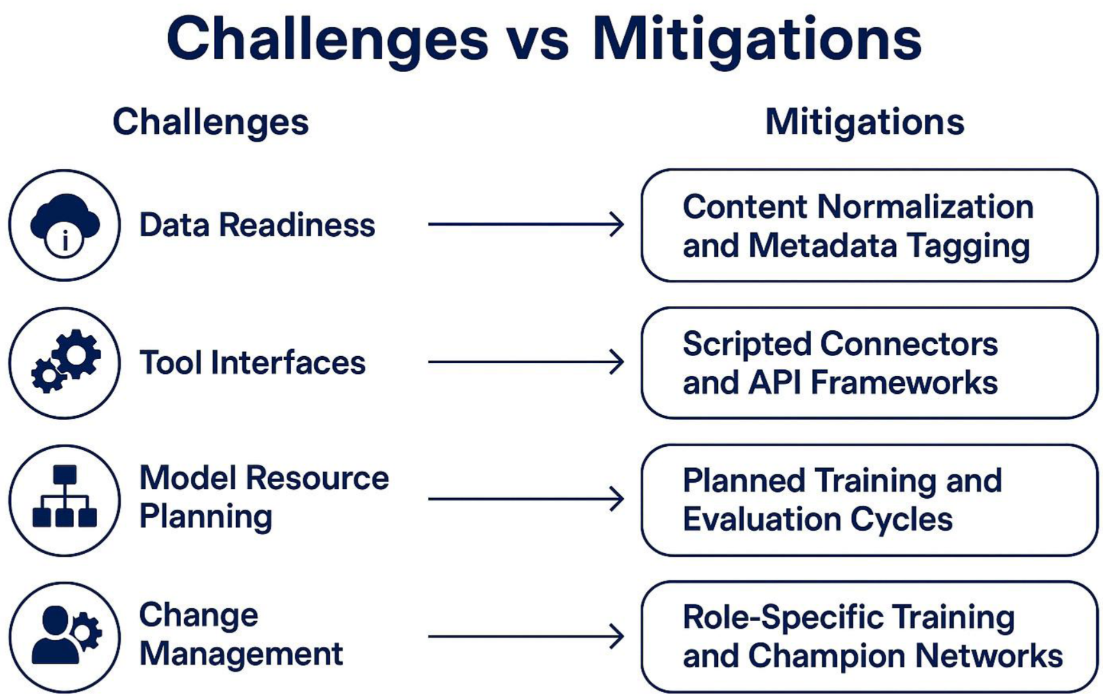
Deliverable	Baseline (hrs)	AI-ED (hrs)	Delta (hrs)	Reduction (%)
P&ID Development	8-9	2 - 3	5-6	65-75
Datasheet review	2-3	0.5 - 1	1-2	50 %

POC Outcomes

PoC activities indicate measurable optimization in engineering deliverable development effort for the tested scopes, coupled with a tangible potential for schedule compression through accelerated engineering cycles. These gains are not incidental; they stem from a deliberate strategy of procedural scoping, which focuses on high-volume, standards-intensive tasks where AI can deliver repeatable value without compromising engineering authority. By embedding standards-aware prompts, the system ensures that generated outputs adhere to ADNOC Group Engineering Standards, thereby reducing rework and minimizing the risk of non-conformance. Furthermore, the use of structured input forms significantly reduces variability in data capture, enabling deterministic rule checks and improving first-time-right performance. Integration with incumbent CAD/CAE and document control systems eliminates format friction and redundant manual steps, creating a seamless flow of information across the tool chain.

However, these benefits are contingent upon overcoming several challenges. Chief among them is data readiness, which requires rigorous content normalization, metadata enrichment, and taxonomy alignment to ensure that AI models operate on clean, structured inputs. Tool interface complexity presents another barrier, necessitating the development of robust APIs and scripted connectors to maintain interoperability across heterogeneous platforms. Model resource planning, including compute allocation, training datasets, and evaluation pipelines—must be addressed early to avoid bottlenecks during scale-up. Finally, change management remains critical: successful adoption hinges on role-based enablement for SMEs, clear governance frameworks, and cultural alignment to position AI as an assistive, auditable tool rather than a disruptive force.

Mitigation strategies are multi-pronged. On the data front, initiatives such as content normalization and metadata tagging establish the foundation for reliable retrieval and standards enforcement. Technical integration risks are mitigated through scripted connectors and API frameworks that enable bidirectional data exchange between AI services and engineering applications. To ensure model robustness, organizations should implement planned training and evaluation cycles within an MLOps framework, incorporating drift detection and benchmark testing. Finally, role-specific training programs and champion networks foster user confidence, while structured feedback loops feed operational insights back into model refinement and governance updates.



Conclusions

1. **AI is practical and impactful when applied to procedural tasks that are well-defined, standards-driven, and supported by structured inputs**. Rather than attempting to automate entire engineering workflows, the most sustainable value emerges from targeting high-volume, repeatable activities such as datasheet preparation, P&ID annotation checks, and standards compliance verification. These tasks lend themselves to AI augmentation because they operate within clear logic boundaries and established rule sets, allowing outputs to be both auditable and traceable. By embedding standards-aware prompts and leveraging structured input forms, organizations can ensure that AI-generated content aligns with ADNOC Group Engineering Standards and remains under the governance of subject matter experts.
2. **Early Proof-of-Concept (PoC) initiatives demonstrate tangible benefits in both cost and schedule performance, validating the business case for AI-assisted engineering design**. POC results indicate an approximate **10% reduction in engineering deliverable development effort** for selected scopes, alongside a potential **2–4 month compression of project schedules**. These gains are primarily driven by reduced rework, faster first-pass drafting, and the elimination of redundant manual steps through seamless integration with CAD/CAE and document control systems. However, these benefits are not guaranteed; they are contingent upon the maturity of data ecosystems, the robustness of integration frameworks, and the readiness of engineering teams to adopt AI-enabled workflows.
3. **Long-term success depends on a disciplined approach to data governance, interoperability, and capability building**. Data standardization—through content normalization, metadata enrichment, and taxonomy alignment—forms the foundation for reliable AI performance. Equally critical is the development of robust interfaces and APIs that enable smooth interoperability with incumbent CAD/CAE and simulation platforms, ensuring a single source of truth across the engineering toolchain. Finally, organizations must invest in model lifecycle management, including training, evaluation, and drift monitoring, supported by an MLOps framework. Complementing these technical enablers, structured change management programs, role-based training, and the establishment of champion

networks are essential to foster user confidence and embed AI as an assistive, auditable tool rather than a disruptive force.

In summary, AI in engineering design is no longer a distant aspiration but an actionable reality —provided it is deployed with precision, governance, and a clear focus on procedural value. Organizations that embrace this paradigm shift with a structured roadmap, robust data foundations, and strong stakeholder engagement will be well-positioned to achieve measurable improvements in cost efficiency, schedule certainty, and overall design quality.

Acknowledgments

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Nomenclature

AI – Artificial Intelligence
LLM – Large Language Model
CAD – Computer-Aided Design
CAE – Computer-Aided Engineering
PoC – Proof of Concept